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Chaos 101

Project Phase 2: Specifications

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1. Excellence

1.1. Objective

Introduction to chaos theory

It is common to think about the world as composed of two main types of systems. On the one hand, there are deterministic systems, such as a simple pendulum or planetary orbits. Their future states can be predicted with a high degree of accuracy once the governing laws and initial conditions are known. On the other hand, there are random systems, such as thermal fluctuations or the radioactive decay of an atom, which behave in an unpredictable manner and can only be described in terms of probability. Beyond these two main categories there is actually another class of systems whose evolution in time is perfectly described by some rules, meaning it is perfectly deterministic, yet turns out to be completely unpredictable over the long term: this is called chaos. This inherent unpredictability arises from the system's extreme sensitivity to initial conditions. If two identical chaotic systems are initiated from states that differ by only a microscopic margin, their trajectories will eventually yield completely different outputs. This phenomenon is widely known as the *butterfly effect*, a concept first formally described by Edward Lorenz during his study of weather models in 1963 [1]. Since those foundational studies, chaos theory has developed over the decades. Researchers have identified and described chaotic dynamics in numerous natural and artificial systems, namely weather patterns, chemical reactions, lasers, fluid dynamics and electronic circuits [2]. Over time, chaos has also increasingly been studied with an engineering approach to be used for real-world technology. Several research fields are nowadays being explored, like cryptography and neuronal modeling [3, 4], but also concrete applications such as the True Random Number Generators (TRNGs) have emerged, making chaos theory a highly relevant field of study in contemporary technological research [5, 6].

The electronic approach

In the context of chaos theory, electronic circuits are of huge interest. This branch of research has long explored the possibility of implementing chaotic behavior within electronic systems, starting with the pioneering work of Leon Chua [7]. The electrical domain is particularly advantageous because it provides a large degree of freedom for modelling differential equations and integrating non-linear components. Moreover, electronic circuits are robust, controllable, easily measurable, and characterized by a fast response time. For these reasons, chaotic circuits have been extensively studied and represent the main platform for investigating chaotic behaviour [8, 9].

The educational problem

Despite its prevalence in nature and engineering, chaos theory remains a challenging subject to comprehend and communicate, mainly due to the inherent difficulty in understanding and describing these complex, non-linear dynamics. Taught through abstract mathematical models, students and general audiences consistently struggle to visualize chaotic behavior and grasp its physical meaning. This significant educational gap leaves people largely unaware of what chaos is and how its principles apply to real applications and become useful.

Project objective

To address this significant educational gap, the absolute priority of this project is to **build a tangible educational device** easily understandable for the general public that aims to provide the answers of two fundamental questions:

"What is chaos?" and "Why is it useful?"

The project aims to provide an intuitive and visual comprehension of a chaotic system, through an electronic circuit exploiting its several advantages. The aim is to illustrate the fundamental properties of chaos and its behaviour, and to provide a direct demonstration of a concrete application. This device should be portable, affordable, interactive and the best alternative to traditional (and sometimes expensive) lab equipment.

Expected outcomes and target audience

The expected outcome of this project is to deliver a functional educational package that includes a real working device and the associated documentation. Moreover, it aims to reach a highly diverse audience. The primary intention is to engage people attending science fairs: non-experts, students, children, but also stimulate researchers. This educational package is also envisioned to be utilized in other educational channels, in particular high schools or universities, as a kit for laboratory activities. It might serve as an interesting lab project to spark curiosity or, even more, as a useful practical tool to study non-linear dynamical systems.

1.2. Concept

The project idea

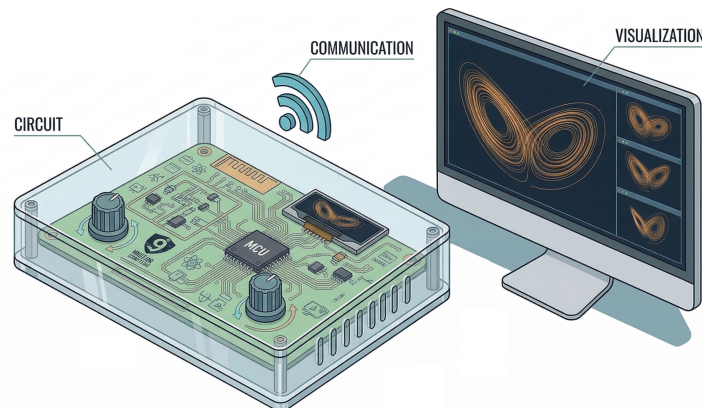


Figure 1: Project concept. Generated with AI Gemini

As presented, the goal of the project is to develop a device which will provide a laboratory experience related to chaotic systems. This device is based on a chaotic circuit, due to the numerous aforementioned advantages of working with electronics. Given its educational character, the name for this device and the entire project came naturally: CHAOS 101. The overall idea is to couple a visualization system to the chaotic circuit to display in real time its dynamical evolution and implement several features of interaction. The first question and the primary message to convey is “*What is chaos?*”. To tackle this, the main functions and features envisioned to be progressively implemented are:

1. **Visualization system:** direct monitoring of the system’s evolution and display of the characteristic phase space diagrams, called *attractors*. This is the most powerful aspect of the project, as it allows us to explain the order hidden behind chaos. It is envisioned to be developed through a microcontroller interface and wireless communication from the device to the visualization system. This allows for a simple setup procedure and opens the possibility of smartphone connection, which might be very useful for science fairs.
2. **Transition from periodic dynamics to chaos:** control over system parameters to transition from periodic to chaotic behavior. This allows for a hands-on experience where users can appreciate the transition and understand that it is something controllable and does not arise randomly.
3. **Slowed-down evolution:** system modifications to allow a slower evolution, demonstrating the phase space diagram formation and conveying a clear intuition of the unpredictable evolution.
4. **Overlapping of trajectories:** overlapping system trajectories with different initial conditions on the visualization, to show the sensitivity to initial conditions. This serves to convey the core concept of the definition of chaos.
5. **Open-source access:** an open-source repository to access all the useful material. The aim is to extend the project’s impact beyond the live presentation provided by the operator. This will include the full technical documentation and a guide to set up the experience, making CHAOS 101 accessible autonomously to everyone.

The second main question to address is “*Why is it useful?*” and for this the last feature to implement is:

6. **Practical application:** demo of a real application to transfer the relevance of chaos theory, in particular chaotic circuits, in technology and research.

Selection of the application

As previously mentioned, several research fields and concrete applications of chaos exist; thus, different options were analyzed to select the most suitable one. A significant technology in this context is the synchronization of multiple chaotic circuits. This phenomenon is the fundamental working principle of chaos-based encrypted communications, and it has also been widely studied as a physical model in biomedical engineering to simulate neuronal activity [10, 11, 12]. This is particularly relevant in medical research concerning Parkinson’s disease and the effects of Deep Brain Stimulation (DBS). However, the hardware implementation of synchronized circuits proved to be too elaborate within the scope of this project. Beyond the other applications, one revealed to be equally interesting and actually feasible: the True Random Number Generators.

Application: True Random Number Generator (TRNG)

TRNGs are a leading technology in the field of information security, where random number generation (RNG) is a key step to ensure high-security encryption. The TRNG represents the highest quality of RNG, performing over the widespread pseudo-RNG whose limitations are extensively known. The main feature of a TRNG is the coupling with a random physical process which guarantees a higher unpredictability. Common sources include thermal noise, phase jitter in oscillators, and also chaotic systems [6]. While the first two approaches are often affected by non-ideal circuit and environmental conditions, chaotic circuits are a perfect candidate for this application, leveraging on their nonperiodicity, broadband spectrum and their high sensitivity to initial conditions. The research is currently strongly focusing on optimizing chaotic TRNGs for ultra-compact, high-throughput hardware architectures, working on FPGAs and CMOS technologies [13]. Due to their superior energy efficiency, these modern chaotic generators are actively proposed for securing resource-constrained environments, such as Internet of Things (IoT) ecosystems, biomedical sensor networks, and edge computing devices[14].

1.3. Approach

Towards the technical implementation of the project several initial aspects need to be elaborated, in particular the feasibility assessment of hardware, TRNG implementation, communication plan and market analysis.

Feasibility assessment of hardware

A first investigation aims to select the most suitable chaotic circuit to implement. Specifically, the evaluation focuses on well-documented designs: the Chua's circuit and its variants and the Sprott Jerk circuit [15, 16, 17]. The system of choice should balance manageable technical complexity with a high visual impact, allowing for the visualization and evolution of the "scenographic" shape of the attractor. The chaotic oscillators are evaluated based on their complexity, the availability of components in the university laboratory, and the shape of their phase space diagrams. Simulations are conducted using KiCAD and LTspice to gain familiarity before the physical implementation. This ensures that the chosen models are manageable and promising before moving to hardware implementation. After a comprehensive study of various chaotic oscillators, the selected circuits are tested in laboratory setup on the **breadboard** to validate the robustness of the design, the correct functioning, and reasonable controllability. Concerning the visualization the main interest is to develop a wireless communication between the hardware and the visualization. To this aim an interface using a **microcontroller** and **Wifi protocol** for the communication seems to be the most promising solution opening the possibility of smartphone connections. Initial testing is performed in the lab to assess its suitability.

TRNG implementation

To implement a reliable TRNG, the two main aspects to explore are the random number extraction and the validation tests. For the extraction, the main techniques that are considered are reported in Table 1. Concerning the validation, the methodology includes randomness and statistical testing. According to standards, the required validation test to be performed is the NIST suite [18]. However, in line with the pedagogical character of CHAOS 101, the testing is intended to be displayed in real-time. For this reason the full NIST is not feasible. Thus, alternative less formal tests have been identified, reported in Table 2.

A further idea is to show the superiority of the intrinsic entropy of hardware chaos with respect to the deterministic software algorithms used in pseudo-RNGs. To make this superiority tangible in terms of unpredictability and security, CHAOS 101 plans to showcase a side-by-side comparison between the circuit-generated sequence and a computer-generated one, highlighting how the proposed hardware system leverages sensitivity to initial conditions at a physical level so that even micro-environmental variations make the future sequence mathematically impossible to predict with precision, providing higher safety.

Extraction	Working Principle
ADC & LSB	Periodic sampling retaining only least significant bits [6].
Thresholding	Analog comparison against a reference voltage to directly generate binary pulses [6].
Poincaré	Event-driven sampling triggered exclusively by specific state-space crossings [6].

Table 1: RNG strategies

Validation	Objective / Metric
Entropy	Evaluates information density [19].
χ^2 Test	Determines how much empirical distribution deviates from uniform [20].
Monte Carlo	Spatial test mapping to (x, y) ; precision of π calculation depends on randomness quality [20].

Table 2: Statistical tests for TRNG

Communication plan

In order to showcase CHAOS 101 in a captivating and effective way, a communication plan will be developed. This will be designed for the audience to best experience chaos, considering not only what explanations to give, but also how to best convey them, for example, through **direct interaction** with CHAOS 101 or using a smartphone connection. To this end, a **guide** for the lab activity will be created, and these interactive aspects will be considered. In addition, other resources, such as posters to display during the events, may be produced. Furthermore, all documentation and materials will be carefully organized and **shared** through an **open-source platform** as a resource to autonomously develop the lab project.

Market Analysis

To validate the project's assumptions regarding the general lack of awareness of chaos theory, an initial survey is going to be conducted targeting the general public, in particular students, and friends and relatives which still represents a good statistical set. The aim is to better solidify the relevance of the project.

1.4. Ambition

CHAOS 101 is not alone in the context of educational projects; there are already numerous useful resources available online. These have been analyzed, to identify where this project can bring significant improvements and distinguish itself from existing solutions.

Analysis of competitors

The most common methods to illustrate chaotic behavior can be broadly classified into two groups: purely digital simulations and purely physical systems. On the one hand, **digital methods** are simple to implement in classrooms and require little hardware, such as interactive simulations offered through websites or software tools like MATLAB or Python-based visualizations. They enable users to change system parameters and see the behavior that results in real time. These methods, however, don't involve **any real interaction**. By using peripheral devices like a mouse or keyboard to abstract user input, the intuitive, physical understanding of the system is reduced.

On the other hand, **physical demonstrations** such as the Double Pendulum or a Magnetic Pendulum provide a direct, hands-on experience. These systems effectively illustrate sensitivity to initial conditions and are visually compelling. However, they are inherently limited in terms of controllability and parameter variation. Typically, the only form of interaction is the initial displacement, after which the system evolves autonomously.

Concerning electronic projects, several educational resources already exist covering all the different design circuits, Chua's, Jerk's, Lorentz's to mention the most important. These contents provide documentation, tutorials, and video explanations, but these are generally self-contained in a website in the context of a "**Do-It-Yourself**" project and not widely available as a fully integrated, user-friendly educational device. They must be conducted in a lab setup, with costly instruments and they are not rapid and easy to assemble.

Envisaged innovation

CHAOS 101 aims to address all of these limitations. It aims to provide the double approach combining both physical interaction with dynamic parameter control. In contrast to purely mechanical systems, the circuit provides a way to adjust parameters, enabling transition from periodic to chaotic behavior.

In addition, compared to existing electronic projects, CHAOS 101 brings key innovations:

- **Fully integrated physical interface**, allowing for easy interaction with the circuit.
- **Autonomous supply**, making it a portable device
- **Wireless data transmission system**, allowing for simple setup and direct streaming of the state variables evolution into a web interface, eliminating the need for additional measurement tools such as an oscilloscope.
- **Digital visualization** opening numerous possibilities to share the resources with people rapidly and effectively.

Overall, it is worth noting that although the basic idea of an electronic chaotic circuit is not a new one, the contribution of this project lies in the integration of interactivity, accessibility, and modern connectivity. This makes the system particularly well-suited for educational and demonstrative purposes, where existing solutions either lack physical engagement or sufficient flexibility.

2. Impact

2.1. User Needs Analysis

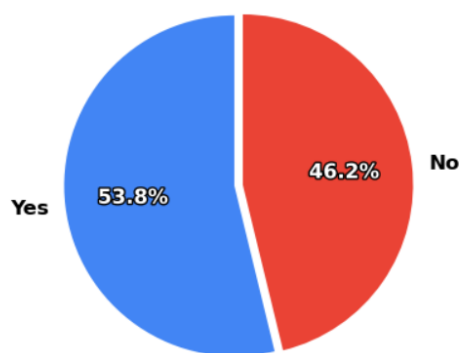
To understand more precisely the impact and the relevance of CHAOS 101, a survey on chaos theory was shared with friends, students and relatives, who form a representative statistical set of the target audience. The goal was to assess the current level of awareness of the general public and bring foundation to this project. **93 answers** were collected and the results are presented below.

2.2. Survey Results

The survey results reveal a generally **limited familiarity** with chaos theory.

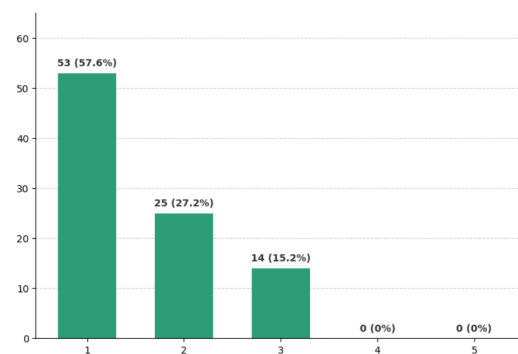
Awareness and Knowledge Level

Have you ever heard about chaos theory?
(93 responses)



(a) Awareness of chaos theory

How much do you know about chaos theory from 1 to 5?
(92 responses)



(b) Self-assessed knowledge level

Figure 2: General awareness and knowledge of chaos theory

As illustrated in Fig. 2a, more than half of the respondents (53.8%) reported having heard of chaos theory, while 46.2% had no prior exposure. However, awareness does not correspond to understanding. In fact, the majority of participants (57.6%) rated their knowledge at the lowest level (1 out of 5), while no respondents selected the highest levels, as shown in Fig. 2b. This shows that even among those who have heard of the term, the actual level of understanding is still very limited. This gap is likely due to the abstract nature of the topic, which is often introduced through mathematical formulations that are not easily accessible to a general audience.

Understanding and Perceived Applications

Chaos is:
(92 responses)

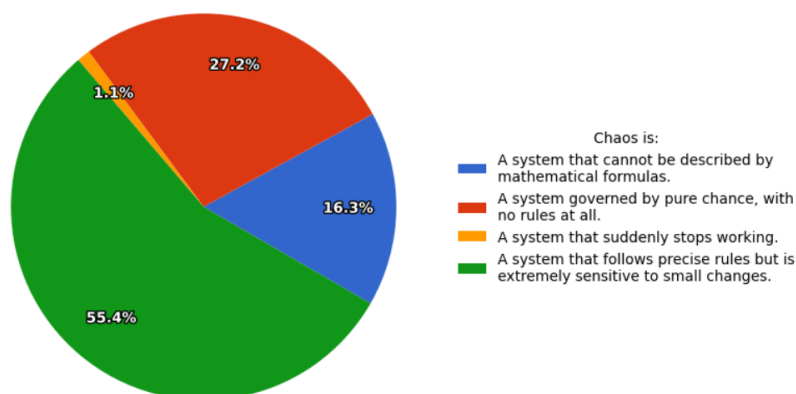


Figure 3: Definition of chaos

Exploitation of chaos:
(92 responses)

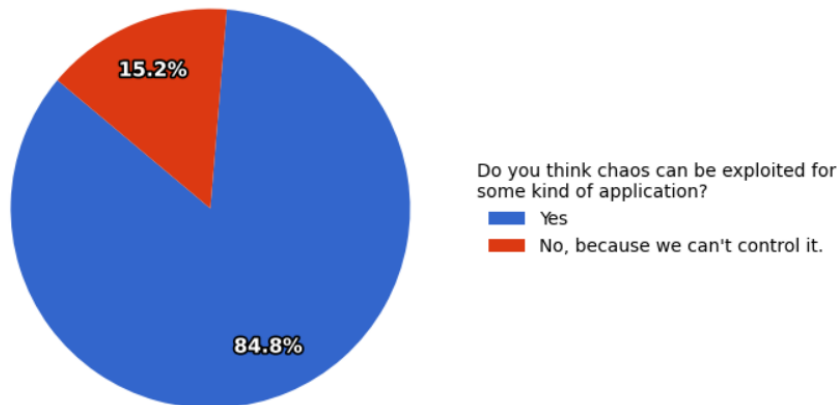


Figure 4: Perceived applications of chaos theory

As illustrated in the plot in Fig.3, when asked about the definition of chaos, only a portion of respondents correctly identified it as a deterministic system highly sensitive to initial conditions. A large number of participants instead associated chaos with randomness or unpredictability, highlighting several common **misconceptions**. Despite this limited understanding, a strong majority (84.8%) believe that chaos can have practical applications, as shown in the plot in Fig. 4. This suggests that people intuitively see that chaos is relevant in real-world phenomena, which could be used as a starting point for educational activities.

2.3. Target Audience

The project is designed for a broad and heterogeneous audience. Its primary target includes **high school and undergraduate students**, with limited exposure to advanced mathematical formalism, who can benefit from an intuitive and visual introduction to complex systems and chaos theory. Additionally, CHAOS 101 aims to reach a **broader public** of all ages, relying on the direct interaction, the visual impact and focusing the activity on a more basic level. The platform is also relevant for **educators** and science communicators, who may adopt it as a demonstrative tool to complement theoretical lessons with interactive experiences. Finally, CHAOS 101 also aims to stimulate **researchers** and highly qualified STEM professionals by showing concrete applications and highlighting the relevance of this research field.

2.4. Educational channels

Among the envisioned educational channels the main targets are **science fairs** and events; CHAOS 101 is intended as an assistive tool for researchers of this field to capture the attention of visitors and provide good communication. As already mentioned, this project is also intended to be used in **schools** and **universities**, as well as in museums and public exhibitions. Finally, internet is another relevant educational channel that will be utilized, particularly through an **open-source platform**, to reach the largest amount of people.

2.5. Market Positioning and Innovation Strategy

CHAOS 101 targets a growing niche market within the **Educational Technology (EdTech)** and science museum sectors. The main innovation is the shift from purely digital simulations to a tangible, hardware-based experience. While the physical implementation of the chaotic system is cost-effective and relatively simple, it offers clear advantages compared to existing software-based alternatives. For institutional users, such as schools and science festivals, the economic benefit is clear: a low-cost, durable device that provides a high-impact, "hands-on" learning experience. A key example of this added value is the integration of the Random Number Generator. By demonstrating a **functional application** in cybersecurity and data encryption, the exhibit is no longer just a visual demonstration, but also a practical proof-of-concept for real-world applications.

3. Implementation

3.1. Project description

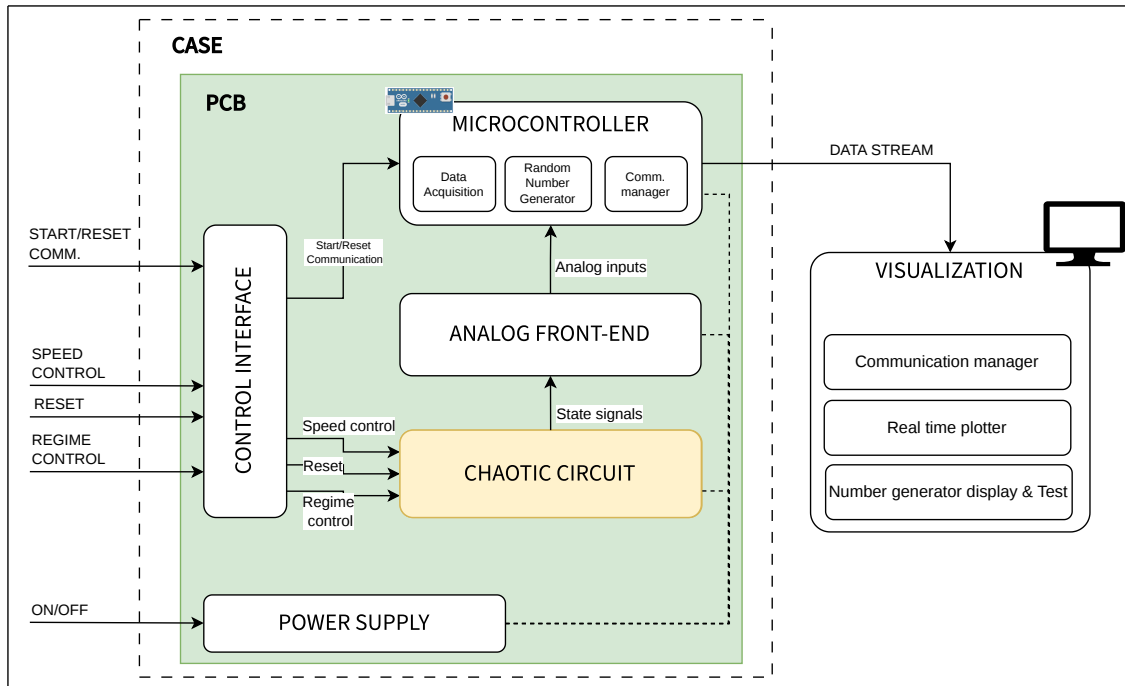


Figure 5: Functional diagram of the project

A general description, shown in Figure 5, is presented regarding how the system is envisioned to be composed, along with the interaction of its main components. CHAOS 101 is composed of a hardware station that hosts the **chaotic circuit** and the electronic system to perform the readout. It features **physical controls** that allow the user to directly modify the circuit parameters and observe different dynamical regimes. The readout is handled with a **microcontroller**, which is properly interfaced to the circuit via an **analog front-end**. The microcontroller processes the acquired data, performs the **random number generation** and transmits everything to the **visualization system** using the most feasible communication protocol (unidirectional communication from micro to visualization). The visualization system then receives the data and produces the graphic display. The hardware station is implemented on a custom PCB and housed in a **protective case**. The entire system is also featured by an autonomous **power supply** to ensure maximum portability for the events.

3.2. Work packages definition

Based on the following structure, the development process is organized into distinct subtasks, referred to as Work Packages (WPs). The identified WP are presented in Table 3.

WP	Denomination
WP1	Chaotic Circuit Development
WP2	Analog Front End
WP3	Firmware for Data Acquisition & Communication
WP4	Visualization System
WP5	Control functions
WP6	Power Supply
WP7	Random Number Generator
WP8	PCB Design & Assembling
WP9	Case Development
WP10	Communication Plan & Branding

Table 3: List of Work Packages.

3.3. Work packages description

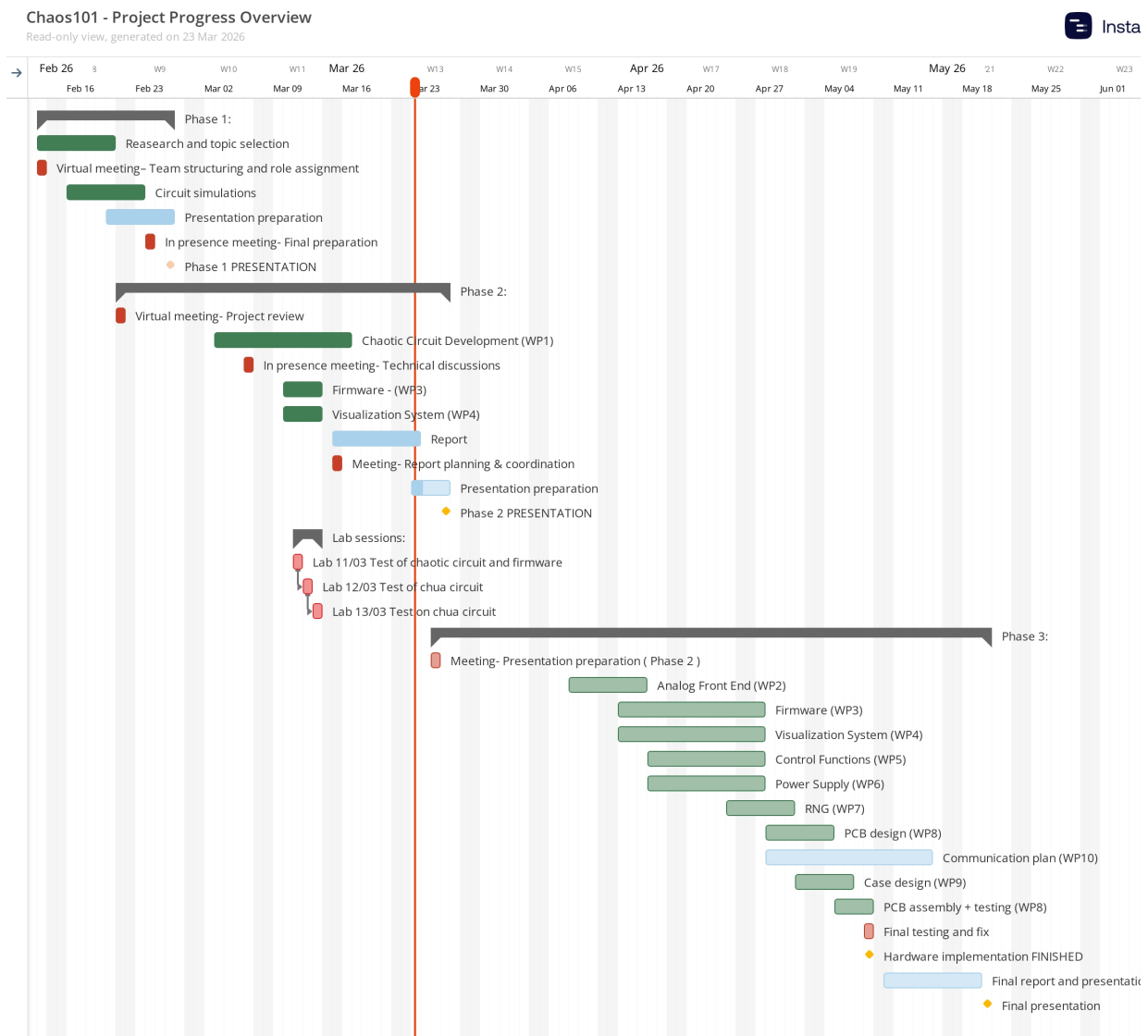
The development process is structured into 10 distinct WPs, each with clearly defined objectives, deliverables and the team resources allocated.

- **WP1: Chaotic Circuit Development** Identification of the chaotic circuit to implement. It involves research in literature, identification of 3/4 circuits of interest followed by simulations using LTspice to study the dynamic and the setup parameters. Following these simulations, the breadboard implementation and physical validation are carried out.
Deliverable: Validated chaotic circuit schematic and assembled circuit onto the breadboard.
- **WP2: Analog Front End** Design, assembling and testing of the analog front end for output signal conditioning from the chaotic circuit to the microcontroller.
Deliverable: Circuit schematic and assembled circuit on the breadboard
- **WP3: Firmware for Data Acquisition and Communication.** Selection of an appropriate microcontroller, and firmware implementation, in particular reading of analog inputs, data processing, and real-time data streaming to the visualization system.
Deliverables: Firmware code
- **WP4: Visualization System.** Development of a local web interface designed to receive data from the microcontroller and plot the chaotic attractors in real-time, offering multiple visualization options for the audience.
Deliverables: Web interface source code
- **WP5: Control functions .** Developing the user interface for physical interaction with the chaotic circuit. It includes integrating variable potentiometers, capacitors to slow down the signal, and control switches.
Deliverables: updated circuit schematics integrating user controls and updated circuit on breadboard.
- **WP6: Power Supply.** This task is dedicated to designing an autonomous power supply system, including battery selection and the implementation of appropriate voltage converters for the microcontroller and operational amplifiers.
Deliverables: Power supply schematic, selected battery specifications and additional components.
- **WP7: Random Number Generator.** Theoretical development and practical implementation of the True Random Number Generator. Design of number extraction at the firmware/hardware level and display features on the visualization system.
Deliverables: updated microcontroller firmware/circuit schematic and updated web interface source code
- **WP8: PCB Design & assembling.** Transferring of the validated schematic circuit, fully tested on breadboard, into a permanent, robust PCB design. Assembling of the PCB and testing, design reiteration if needed.
Deliverables: Fully printed, assembled, and functional PCB.
- **WP9: Case Integration.** Designing and manufacturing of the custom housing for the PCB to ensure safety, robustness portability, and a professional appearance.
Deliverables: Designed and 3D-printed enclosure.
- **WP10: Communication Plan & Branding.** Educational strategy, ensuring the final product effectively communicates chaos theory to the public. It includes defining the visitor experience and creating accessible documentation to share, identify a proper open source platform and share the material.
Deliverables: Support resources for CHAOS 101 and online upload of materials.

WP	Work Package Name	Responsible members
WP1	Chaotic Circuit Development	Giacomo, Finn, Tasnim, Yanis, Matthieu, Diana, Tommaso
WP2	Analog Front End	Matthieu, Tasnim
WP3	Firmware	Anhelina
WP4	Visualization System	Anhelina, Andrea
WP5	Control functions	Yanis, Giacomo
WP6	Power Supply	Finn
WP7	Random Number Generator	Tommaso, Diana
WP8	PCB Design & assembling	Yanis, Matthieu
WP9	Case Integration	Andrea
WP10	Communication Plan & Branding	Giacomo, Tommaso, Diana, Tasnim

Table 4: List of Work Packages.

3.4. Work timeline



The project work is divided into three main phases: the preparation phase, the specification phase (currently ongoing), and the development phase. The key milestones that dictate the project's evolution are the presentations scheduled for:

- **PHASE 1: February 26 – State of the Art:** Bibliographic study, physical principles, and context.
- **PHASE 2: March 26 – Specifications:** Building blocks, project scheduling, and market.
- **PHASE 3: May 19 – Development:** Technical achievements, final report, and discussion.

The work involves laboratory sessions, the scheduling of which is critical for progress. In Phase 2, laboratory sessions were carried out for the feasibility assessment and development of the chaotic circuit (WP1). In the upcoming development phase, the work packages have been allocated appropriate timeframes, and subsequent laboratory sessions will be scheduled for them. Work packages WP3 through WP6 will be carried out in parallel to maximize teamwork and efficiency.

Remark on Meetings Organization It is important to note that weekly meetings are not systematically planned in advance. Due to the collaborative nature of the project, frequent informal discussions take place among team members, with or without the tutor present, often during breaks or working sessions. Formal meetings are organized only when necessary, particularly at key stages of the project, such as for decision-making, progress evaluation, or the preparation of deliverables. This flexible approach allows the team to maximize the effectiveness of meetings.

3.5. Management structure and procedures

The management structure of CHAOS 101 is designed to be highly collaborative and agile. Rather than operating under a rigid hierarchy, for the crucial choices the team relies on a **democratic decision-making process**, thanks to the **active contribution** of each member. This structure ensures all team members are aligned and, more importantly, invested in the outcome. To ensure smooth progress, the **specific decisions** involved in the development are taken by the team members responsible of the work. For the internal communication, the team holds frequent coordination meetings, utilizing both in-person sessions and video conferencing.

While the team operates horizontally, one member actively assumes the role of a **manager**. This role involves maintaining an updated overview of the project, managing work divisions and internal communication, tracking task completion across different work packages, and ensuring that deadlines are met without overriding the democratic consensus of the group on last minute decisions. Furthermore, the team is split into **3 subdivisions**: electronics, research, software as shown in Figure 7. This ensures clear subdivision of responsibilities; however, a strong flexibility is required to fulfil the various tasks, especially in light of project presentations, specifically, drafting reports, and conducting theoretical research.

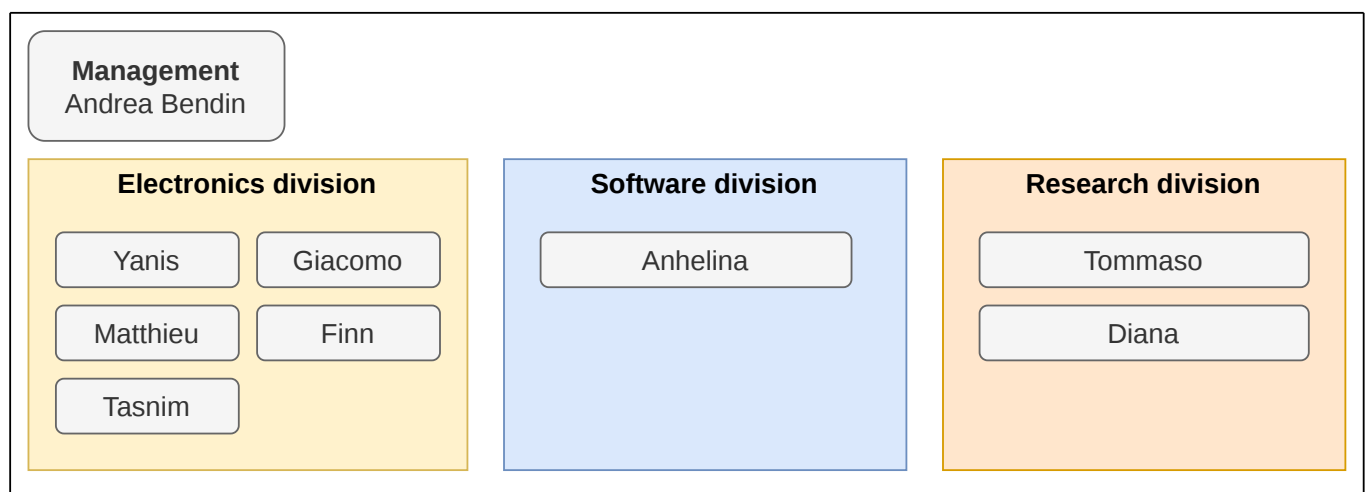


Figure 7: Team structure

3.6. Consortium as a whole

The consortium consists of nine dedicated members who possess a diverse and complementary skill set, allowing for an effective division of labor across specialized sub-groups while maintaining strong overall cohesion.

Yanis, Tasnim, and Matthieu possess relevant skills in PCB design and electronic assembly; therefore, they are responsible for the electronic development, supported by Giacomo, physics engineer, and Finn, electronic engineer, in the assembly and testing phases. Anhelina, biomedical engineer, having previous expertise in embedded system programming, is responsible for the firmware and the visualization system. Diana, physics engineer, and Tommaso, physicist, are in charge of the RNG implementation, leveraging their analytical and technical skills. Andrea, with experience in electronic and mechanical design, acts as the project manager, supervising the overall work of the subdivisions and designing the case.

Division	Member	Expertise
Electronic	Jeljal Yanis	Electronics, PCB design
	Gotas Matthieu	Electronics, PCB design.
	Jalel Tasnim	Electronics, PCB design.
	Tannenberg Finn	Electronics, Optics
	Anfossi Giacomo	Physics, research analysis, electronics.
Research	Lanfranchi Tommaso	Theoretical physics
	Fugnanesi Diana	Physics, electronics.
Software	Kovalchuk Anhelina	Telecommunications, software development.
Management	Bendin Andrea	Electronics, mechanical design.

Table 5: Consortium members grouped by operational division and specialized expertise.

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